DATA INVENTORY · COMPILED 6 SEPTEMBER 2026

The datasets

Every data source behind the empirical evaluation of NIST's LEV metric — what was used, at what version, for which result, and what each one cannot tell us. Plus the six further sources identified for the next phase.

These are analysis inputs, not training data. No model is fitted to any of them — see "why nothing is trained" below.

5	14.68 M	380,880	1.8 GB
datasets used	records ingested	distinct CVEs touched	on disk, 238 EPSS files

PART 1

What was used

Five sources. Each row gives the exact version, because LEV values change with the analysis date and with the EPSS model version, so "EPSS data" alone would not be reproducible.

SOURCE	EXACT VERSION	RECORDS	ROLE IN THE STUDY
EPSS FIRST / Cyentia epss.empiricalsecurity.com	67 monthly snapshots, 1st of month, 2021-04-14 to 2026-08-01, plus the Table 5 anchor dates. 171 additional daily files for the retraining-boundary probes and the daily-vs-monthly check. 238 files total.	14,456,555 score rows	**The sole input to LEV.** Every LEV value in the study is a product over these scores across 30-day windows.
CISA KEV cisa.gov	catalogVersion 2026.08.25, released 2026-08-25T17:00:12Z. dateAdded spans 2021-11-03 to 2026-08-25. SHA-256 `4ddd6177…3bb79`	1,676 entries	**The only exploitation ground truth.** Defines "observed" in every calibration figure.
CVE List v5 CVE Program	Release `cve_2026-08-25_1700Z`, asset `2026-08-25_all_CVEs_at_midnight.zip.zip`, CVE-2022 onward. SHA-256 `87ca8c2c…51e823`	192,054 CVE records	CVSS v3, CWE and vendor attribution for the top-bucket stratification. 80.5% carry CVSS, 81.9% vendor, 81.4% CWE.
Exploit-DB gitlab.com/exploit-database	Snapshot committed to the repository rather than fetched, because upstream serves it from a moving branch. SHA-256 `54b9167e…df9487`	25,057 CVEs referenced	Public-exploit availability, as one feature separating KEV from non-KEV CVEs at identical LEV.
Metasploit rapid7/metasploit-framework	Module index snapshot, also committed rather than fetched. SHA-256 `c155aa27…3e5ee99`	3,185 CVEs referenced	As above. Combined with Exploit-DB: 26,311 unique CVEs with a public exploit.

Which source produces which result

So a reader can trace any figure back to its inputs.

RESULT	SOURCES USED	HEADLINE OUTPUT
CSWP 41 Table 5 reproduction	EPSS only	All four published values reproduced within 11%
Calibration / reliability	EPSS + CISA KEV	17.2x over-prediction; Brier 0.037, ECE 0.076
EPSS version drift	EPSS only	64–70% of score rise happens at retraining boundaries
Version-normalized LEV	EPSS only	26.9% of LEV mass is version artifact; 17.2x → 12.6x
Analysis-date sweep	EPSS + CISA KEV	Growth is convex, never saturates
Top-bucket stratification	EPSS + KEV + CVE List v5 + Exploit-DB + Metasploit	14x KEV-rate spread by vendor at identical LEV
Vendor-ceiling bound	EPSS + KEV + CVE List v5	Over-prediction bounded at 3.4–4.1x
Sampling-error check	EPSS daily files	Monthly sampling costs 3.9%, unsigned

The population actually analysed

367,311	CVEs with any EPSS data
→ 170,614	first scored in the EPSS v3 era, 2023-03-07 onward
→ 148,920	after excluding CVEs EPSS backfilled retroactively
→ **105,250**	**the calibration cohort** — also at least six months old, so each has had a chance to appear in KEV
495	**of those are in CISA KEV — a base rate of 0.47%**

> **A coverage gap worth stating.** Of the 1,676 CISA KEV entries, **1,665 have EPSS data**. The remaining 11 have no EPSS score at any date and therefore cannot receive a LEV value at all — LEV is structurally silent on them.

PART 2

What each source cannot tell us

The limitations matter more than the counts. Each is documented at the line of code that introduces it; the study records 44 threats to validity in total.

SOURCE	LIMITATION
EPSS	Sampled monthly rather than daily — measured cost 3.9% of mean LEV, and unsigned, so it adds noise not bias. Spans three model retrainings; one tripled corpus scores overnight, and LEV compounds across that boundary as though it were evidence of exploitation.
CISA KEV	**Three problems.** (i) Incomplete by design — CISA lists only what it confirms and prioritises for federal remediation. (ii) `dateAdded` is when CISA catalogued, not when exploitation happened. (iii) It is an input *feature* to EPSS, so it partly predicts the very score being tested against it; removing that leak moves over-prediction from 11.9x to 32.7x.
CVE List v5	Vendor names are unnormalised CNA free text, and a CVE affecting many products is attributed to the first listed. Per-vendor counts are lower bounds.
Exploit-DB **Metasploit**	Record exploit *availability*, not exploitation in the wild — a different and much larger population. Both are also EPSS input features, so they correlate with LEV by construction. Valid as covariates; **not** valid as a second ground truth.

PART 3

Identified for the next phase

Access-checked 6 September 2026. All six are free; four need no account at all.

SOURCE	ACCESS	WHAT IT ADDS	CAVEAT
VulnCheck KEV Community	`free key`	The second exploitation list. ~3.6x CISA's volume, with per-report provenance and dates.	A superset of CISA KEV as shipped, so an independent list must be reconstructed by filtering out CISA-sourced reports.
CIRCL Vulnerability-Lookup	`open`	Third list. Carries Shadowserver honeypot telemetry, no contract needed.	Coverage depth unverified.
ENISA EUVD	`open`	Fourth list, EU-sourced, partially independent.	Its KEV dump is a CISA+EU consolidation; use the exploited-flag search.
GreyNoise Research Program	`apply`	Internet-wide sensor telemetry — the strongest independence claim available.	Reviewed application, so start early. Also an EPSS label source, so partly circular.
CISA KEV daily archive	`open`	Verifies `dateAdded` is never backdated, currently assumed from one snapshot.	None material.
EPSS full daily archive	`open`	Removes the 3.9% sampling approximation entirely. ~2.5 GB.	No licence declared on the repository.

6	11	0	1
identified for the next phase	datasets in total	requiring paid licences	needing a reviewed application

METHOD

Why nothing is trained

LEV is a formula, not a model. Nothing is fitted, so there is nothing to train.

A classifier *could* be built from these sources — 105,250 CVEs with 495 positives. It is not, for four reasons.

Wrong question	The question is whether NIST's published metric is calibrated, not whether a better predictor could be built.
Broken labels	Any model here trains on CISA KEV membership — the thing shown to be incomplete and vendor-biased, 32.5% of Microsoft CVEs against 2.3% of XWiki's at identical LEV. It would learn CISA's visibility, not exploitation.
Circularity	EPSS already uses "listed on CISA KEV" as an input feature; training on KEV labels with EPSS-derived inputs compounds that leak rather than avoiding it.
Weaker by construction	It would reproduce EPSS with less — 1,477 features and four commercial honeypot feeds, against five public sources here.

> **The strongest findings use no labels at all.** That 26.9% of LEV mass is an EPSS version artifact, and that LEV growth is convex rather than saturating, are measured from EPSS alone — which is why the catalogue-incompleteness objection cannot reach them. **The bottleneck is data, not model capacity.**

PROVENANCE

How this stays reproducible

Pinned end date	The EPSS archive end is fixed in code at 2026-08-01. Without it the loader defaults to today, and re-running weeks later silently ingests newer snapshots.
Committed sources	Exploit-DB and Metasploit are served upstream from moving branches, so both snapshots are committed to the repository rather than re-fetched.
Hashes recorded	SHA-256 for every fixed-file source, with release tags, retrieval dates and record counts, in `DATA_SNAPSHOT.md`.
One command	`make fetch` pulls the data once; `make reproduce` rebuilds every table and figure offline, in order.

> **Two licences remain unresolved.** VulnCheck's attribution notice is silent on peer-reviewed publication of derived data and on archiving raw data for reproducibility, and the EPSS history repository declares no licence at all. Both need settling in writing before submission.

Data inventory for the empirical evaluation of NIST CSWP 41 (Mell & Spring, *Likely Exploited Vulnerabilities*, May 2025). Analysis date 2026-08-01. Companions: **LEV-calibration-Phase1.pdf** (findings) and **Phase2-Datasets.pdf** (next-phase sourcing detail). Full provenance in **DATA_SNAPSHOT.md**.